STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-GLUEING

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1650-1662 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 1664-1698, the complete author proof (33 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 17 of queue_examples-defos.json (pb=1425, stmt 1650-1662, proof 1664-1698).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DY1: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1031–1040. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DY7: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1425–1440. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DYX: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1600–1608. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DYW: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1584–1592. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01JB: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1698–1698. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01JA: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1698–1698. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:examples-defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0DYY: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1638–1648. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Situation 0DYY there is an equivalence*

$$\mathcal{D}ef_X = \mathcal{D}ef_{P_1} \times_{\mathcal{D}ef_{P_{12}}} \mathcal{D}ef_{P_2}$$

of deformation categories, see Examples 0DY7 and 0DY1.

Proof. It suffices to show that the functors of Lemma 0DYW define an equivalence

$$\mathcal{D}ef_X \longrightarrow \mathcal{D}ef_{U_1} \times_{\mathcal{D}ef_{U_{12}}} \mathcal{D}ef_{U_2}$$

because then we can apply Lemma 0DYX to translate into rings. To do this we construct a quasi-inverse. Denote $F_i : \mathcal{D}ef_{U_i} \to \mathcal{D}ef_{U_{12}}$ the functor of Lemma 0DYW. An object of the RHS is given by an A in $\mathcal{C}_\Lambda$, objects $(A, V_1) \to (k, U_1)$ and $(A, V_2) \to (k, U_2)$, and a morphism

$$g : F_1(A, V_1) \to F_2(A, V_2)$$

Now $F_i(A, V_i) = (A, V_{i,3-i})$ where $V_{i,3-i} \subset V_i$ is the open subscheme whose base change to k is $U_{12} \subset U_i$. The morphism g defines an isomorphism $V_{1,2} \to V_{2,1}$ of schemes over A compatible with $\mathrm{id} : U_{12} \to U_{12}$ over k. Thus $(\{1, 2\}, V_i, V_{i,3-i}, g, g^{-1})$ is a glueing data as in Schemes, Section 01JA. Let Y be the glueing, see Schemes, Lemma 01JB. Then Y is a scheme over A and the compatibilities mentioned above show that there is a canonical isomorphism $Y \times_{\mathrm{Spec}(A)} \mathrm{Spec}(k) = X$. Thus $(A, Y) \to (k, X)$ is an object of $\mathcal{D}ef_X$. We omit the verification that this construction is a functor and is quasi-inverse to the given one. □